Curse of Irevelant Response: Lower-resource languages lead to lower following rate. The outcomes for the FOLLOWING RATE are depicted in Table 1. When presented with harmful prompts in high-resource languages, the LLM responds with relevant responses. This enhanced response quality is largely attributed to the extensive training data available for these languages, facilitating a deeper and more nuanced understanding when prompted in these languages. Consequently, even when the LLM generates content with harmful undertones, it frequently responds in a manner that helpfully addresses the harmful prompts.

In the following sections, we aim to (1) find whether such two curses still exist in open-sourced LLMs (§3). (2) try to alleviate these two curses through common alignment strategies (§3). (3) trace the origin of these two curses (§4)

3 Does Alignment Training Lift the Curses of Low-resource Languages?

To trace the root cause for the two curses, we study the effect of alignment training with human preference data for the safety and helpfulness of responses, and observe how the resulting language models' behavior changes to malicious prompts in low- vs. high-resource languages. Specifically, we conduct experiments on the HH-RLHF dataset (Bai et al., 2022a). We compare different instruction tuning strategies, such as *supervised fine-tuning* (SFT) and *reinforcement learning from human feedback* (RLHF) (Ouyang et al., 2022). We additionally explore the effect of SFT and RLHF training in multilingual settings, where the instruction tuning data is translated from English into the target languages for SFT or reward model training respectively.

3.1 Multilingual alignment strategies

Multilingual Supervised Fine-tuning (xSFT) Given an instruction-tuning dataset $\mathcal{D}_{l_1}$, which features pairs of prompt and target responses both written in a high-resource language l_1 (e.g., English), we translate the examples into other target high- and low-resource languages in our evaluation $l_{2..n}$. This yields $\{\mathcal{D}_{l_1}, \mathcal{D}_{l_2}, \ldots, \mathcal{D}_{l_n}\}$. We merge all translated data for instruction tuning of the LLM with the following objective.

$$\mathcal{L}(\theta) = \sum_{P,R \in \mathcal{D}} \ell_{clm}(R|P, \theta) \tag{1}$$

where $\mathcal{D}$ is the combined mixture of all translated datasets $\{\mathcal{D}_{l_1}, \mathcal{D}_{l_2}, \ldots, \mathcal{D}_{l_n}\}$, and P,R refer to instances of the harmful prompts and ethical responses in the dataset. ℓ_{clm} denotes the causal language modeling loss.

RLHF via multilingual reward model (xRLHF) To train a multilingual reward model, we start with a human preference dataset $\mathcal{Q}_{l_1} = \{I_i, r_i^+, r_i^-\}_{i=1}^N$ in English. r_i^+ represents the human-preferred response over the less preferred one R_i^-. We translate the prompts and responses into the target low- and high-resource languages $l_{2..n}$, yielding $\{\mathcal{Q}_{l_1}, \mathcal{Q}_{l_2}, \ldots, \mathcal{Q}_{l_n}\}$. Similar to the xSFT case, we combine all translated human preference datasets and use the mixture to train a multilingual reward model. The reward model learning objective is to minimize the ranking loss $\mathcal{L}$ to the learned scalar reward function $\mathcal{R}_\theta$, where σ is the sigmoid function and $I_i \circ r_i^+$ is the concatenation of I_i and r_i^+.

$$\mathcal{L}(\theta) = -\sum \log(\sigma[\mathcal{R}_\theta(I_i \circ r_i^+) - \mathcal{R}_\theta\left(I_i \circ r_i^-\right)]) \tag{2}$$

With the learned multilingual reward model, we apply RLHF on the xSFT trained model. Specifically, we follow the PPO algorithm (Schulman et al., 2017; Ouyang et al., 2022) and maximize the following combined objective function $\mathcal{J}(\phi)$.

$$\begin{aligned}\mathcal{J}(\phi) = \mathbb{E}_{(I,r)\sim\mathcal{D}_{\pi_\phi^{\mathrm{RL}}}}[\mathcal{R}_\theta(I \circ r) - \\ \beta \log(\pi_\phi^{\mathrm{RL}}(r \mid I)/\pi^{\mathrm{xSFT}}(r \mid I))],\end{aligned} \tag{3}$$

where π_ϕ^{RL} is the learned RL policy parameterized by ϕ and initialized from the pretrained xSFT model π^{xSFT}. $\mathcal{D}_{\pi_\phi^{\mathrm{RL}}}$ and $\mathcal{D}_{\text{pre}}$ denotes the RL training and pre-training datasets respectively. The first term encourage the policy π_ϕ^{RL} to generate responses that have higher reward scores. The second term represents a per-token approximated KL reward controlled by coefficient β between π_ϕ^{RL} and π^{SFT} to mitigate over-optimization toward the reward model during RL training. The set of training prompts used in the RL stage is also translated into the target languages, similar to the xSFT case.

3.2 Experimental setup

Benchmarks and methods We use the HH-RLHF dataset Bai et al. (2022b) to train our xSFT and xRLHF models. For evaluation, we used the harmful prompts collected by Zou et al. (2023). We follow the same evaluation metrics HARMFUL RATE and FOLLOWING RATE, as described in §2.

We use LLaMa2-7B as the base model for mono- and multi-lingual SFT and RLHF instruction tuning. In addition, we compare to the official checkpoint of LLaMa2-chat-7B, which is instruction-tuned with RLHF on safety-related examples as part of the training mixture (Touvron et al., 2023)[2]. For simplicity, we refer to this model as CHAT-RLHF. We include our implementation details in Appendix B.

Translator and languages We use NLLB-1.3B (NLLB Team et al., 2022) [3] as the translation model. Here, we select five high-resource and five low-resource languages respectively for our experiments. The five high-resource languages are *English, Simplified Chinese, Spanish, Portuguese, French*. And the low-resource languages are *Hausa, Igbo, Kamba, Halh, Urdu*. We include a more detailed description of the process and prompts used in Appendix A.

3.3 Results on harmful rate

We start by evaluating the base LLaMa-2 model without further alignment training as the baseline. As shown in Table 2, the base LLaMa2 generally generates harmful responses across all languages. Overall, LLaMa2 (BASE) exhibits an average HARMFUL RATE of 77.4% and 80.4% across high and low-resource languages, with only around 3% gap between these two language resource levels.

Model	High (avg.)	Low (avg.)
LLaMa2 (BASE)	77.4	80.4

Table 2: LLaMa2 (BASE) achieves similar HARMFUL RATE(↓, in percentage) on high-resource and low-resource languages.

Reducing HARMFUL RATE is more difficult with low-resource languages. In Table 3, we show the improvements in terms of HARMFUL RATE after alignment training is applied on the base model. Despite all methods (CHAT-RLHF, xRLHF, xSFT) reducing the HARMFUL RATE of the model, we observe a notable gap between their effectiveness on high-resource and low-resource languages.

Specifically: (1) With the official CHAT-RLHF checkpoint, RLHF training results in a substantial 45% reduction in high-resource languages, but the average improvements drop to around 20% for low-resource languages. (2) In our experiments, xSFT leads to a 20% decrease in HARMFUL RATE for high-resource languages. In comparison, we see a less than 7% reduction for low-resource languages. Similarly, xRLHF results in a 14% decrease in the harmful output rate for high-resource languages, compared to zero improvements for low-resource languages.

Aligned Model	High-resource	Low-resource
	Δ_h (base→aligned)	Δ_l (base→aligned)
xSFT	**23.0** (77.4→57.4)	9.8 (80.4→70.6)
xRLHF	**14.4** (77.4→66.0)	2.4 (80.4→78.0)
CHAT-RLHF	**44.8** (77.4→35.6)	23.4 (80.4→57.0)

Table 3: Improvement (Δ, in percentage) of alignment methods on reducing HARMFUL RATE (↓, a **higher** reduction is preferred) of aligned models. The numbers in parentheses mean the performance changes after alignment.

The results suggest that *harmfulness curse* for low-resource languages persists after alignment training. This highlights the difficulty of resolving the curse with typical alignment training methods.

3.4 Results on following rate

As shown in Table 4, the base LLaMa2 model exhibits low FOLLOWING RATE across all languages without further alignment training or instruction tuning. Specifically, LLaMa2 achieves 33.0% FOLLOWING RATE in high-resource languages and 24.8% in low-resource languages. Notably, we already observe a gap between low- vs. high-resource languages in terms of the instruction following capabilities with the base model.

Model	High (avg.)	Low (avg.)
LLaMa2 (BASE)	33.0	24.8

Table 4: LLaMa2 (BASE) achieves comparable FOLLOWING RATE(↑, in percentage) on high-resource and low-resource languages.

Improving FOLLOWING RATE is more difficult with low-resource languages. Similarly, we observe much smaller gains in terms of FOLLOWING RATE when alignment training is applied on the base model. As illustrated in Table 5, while high-resource languages experience consistent boosts

[2]For the LLaMa-2-chat checkpoints, Touvron et al. (2023) did not reveal details on the safety training data used during RLHF, e.g. distribution of languages, source of data.

[3]https://huggingface.co/facebook/nllb-200-1.3B

in FOLLOWING RATE, the improvements for low-resource languages are much smaller.

Aligned Model	High-resource	Low-resource
	Δ_h (base→aligned)	Δ_l (base→aligned)
xSFT	**4.8** (33.0→37.8)	3.4 (24.8→28.2)
xRLHF	**0.8** (33.0→33.8)	−1.2 (24.8→23.6)
CHAT-RLHF	**57.8** (33.0→90.8)	12.0 (24.8→36.8)

Table 5: Improvement (Δ, in percentage) of alignment methods on reducing FOLLOWING RATE (↑, a **higher** improvement is preferred) of the model. The numbers in parentheses mean the performance changes after alignment.

Here, it is worth noting that despite the big improvements from RLHF training of CHAT-RLHF in high-resource languages, we see a much lower improvement rate when we test it on low-resource languages. Apart from the official CHAT-RLHF checkpoint, our alignment training with xRLHF and xSFT does not achieve significant enhancements in FOLLOWING RATE. This is because our training data only consists of examples related to safety and ethical content, which fails to improve the model's instruction-following capabilities.

3.5 Monolingual SFT fails to resolve the curses

We investigate the improvements of monolingual fine-tuning in different languages in reducing HARMFUL RATE, and the results are shown in Figure 2. From the results, we can see that (1) SFT on high-resource language data only provides improvements on high-resource languages. (2) SFT on low-resource language data is not beneficial for high-resource or low-resource languages. As for FOLLOWING RATE, monolingual SFT on the ethical data generally provides limited improvements for enhancing FOLLOWING RATE. This is reasonable since our ethical datasets aim to reduce harmfulness instead of enhancing LLMs' instruction-following or chat ability.

4 Where does the low-resource language curse stem from?

Our earlier experiments (§3) reaffirm the presence of the two curses in open-source LLMs. This is consistent with findings from the GPT-4 experiments (§2). The recurrent patterns suggest that these curses are not mere coincidences, driving us to investigate their origins. For clarity, we break down the LLM training process into two stages: (a) The pre-training stage, where the LLM is trained on a vast corpus using causal language modeling loss. (b) The post-hoc alignment stage, where the pre-trained LLMs are further fine-tuned using alignment data.

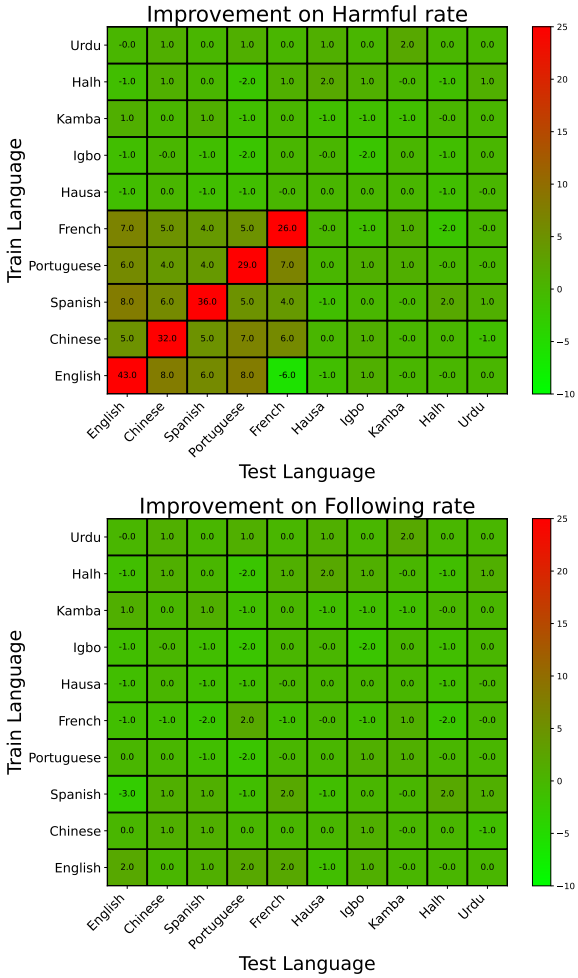

Figure 2: Monolingual SFT fails to improve HARMFUL RATE and FOLLOWING RATE on low-resource languages. The value in the heatmap corresponds to the change of HARMFUL RATE (top figure) and FOLLOWING RATE (bottom figure) after monolingual SFT is applied. Specifically, the red region (in the top figure) represents a large improvement, demonstrating the effectiveness of monolingual SFT on high-resource languages.

Harmfulness curse. LLMs without alignment suffer from malicious prompts, regardless of the language. Based on our results in Table 2 (full results can refer to Table 12), LLaMa2 (BASE) achieves a similar average HARMFUL RATE on low-resource and high-resource languages, and we do not observe any significant bias towards languages from different resource levels.

Instead, as shown in the results of our alignment